

- 6 Fig. 6.1 shows a simple transformer consisting of a primary coil of 25000 turns and a secondary coil of 625 turns.

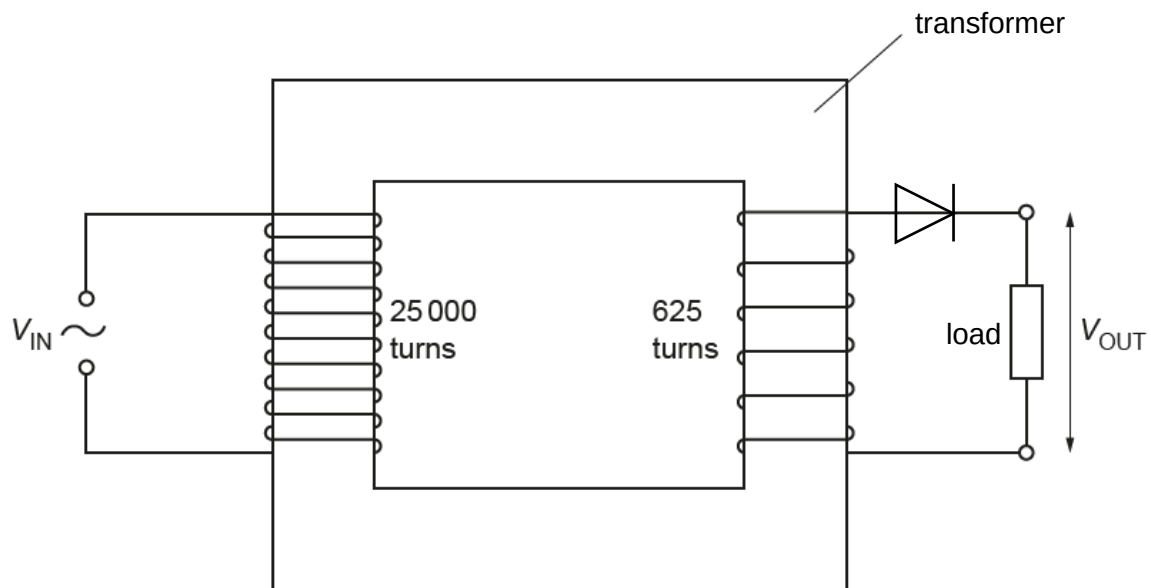


Fig. 6.1

The primary coil is connected to an alternating voltage V_{IN} which is represented by the equation

$$V_{IN} = 220 \sin(120\pi t)$$

The secondary coil is connected to a diode and a load in series. The voltage across the load is V_{OUT} .

- (a) Calculate the r.m.s. value for V_{OUT} .

r.m.s. $V_{OUT} = \dots\dots\dots$ V [3]

$$N_S / N_P = V_S / V_P$$

$$V_{S\text{-peak}} = (625 / 25000) \times 220 \quad \text{OR} \quad V_{S\text{-rms}} = (625 / 25000) \times 220/\sqrt{2}$$

$$\text{rms } V_{OUT} = V_{S\text{-peak}} / 2 \quad \text{OR} \quad \text{rms } V_{OUT} = V_{S\text{-rms}} / \sqrt{2}$$

$$= 2.75 \text{ V (or 2.8 V)}$$

Most students did not seem to realize that the output on the secondary coil is half rectified which means the rms value is half the peak value.

- (b) An electrician mistakenly connected the diode to the primary circuit instead as shown in Fig. 6.2.

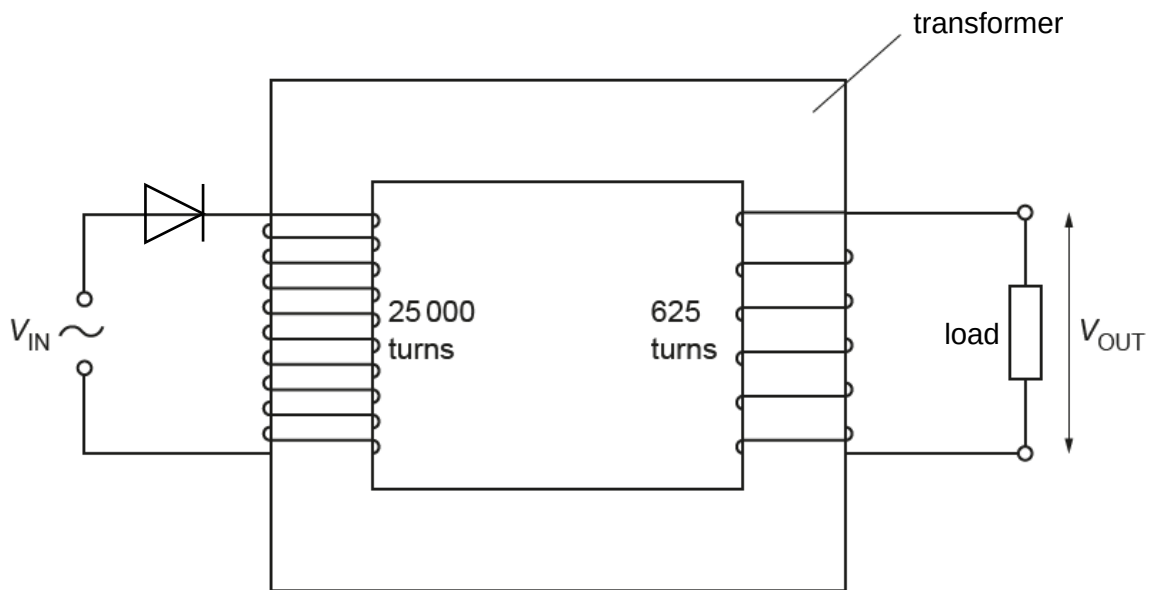


Fig. 6.2

The load requires a direct current to flow in it to operate properly. However, due to the mistake, an alternating current flow in it instead and the load fails to operate properly.

Use Lenz's Law to explain why an alternating current instead of a direct current would flow in the load.

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[3]

(In 1 cycle,) Current in primary coil (in the same direction) increases and decreases leading to increase and decrease in magnetic flux linkage in secondary coil

When magnetic flux linkage increases, current in secondary coil will flow in a direction by Lenz's law (to produce a magnetic flux) that opposes the increase in magnetic flux linkage

When magnetic flux linkage decreases, current in secondary coil will flow in the opposite direction hence alternating

Several students thought the input was alternating which means the whole understanding of the situation is wrong. Others knew that it was half-rectified but thought that the alternating output was due to the sudden switching on and off of the current which is again a wrong understanding.

It is key that responses should state that the magnetic flux increases and decreases. Then Lenz's law should be used to deduce that this will lead to currents of opposite directions.

Many responses were generic. They quoted and used the laws correctly but did not use them in the context of the question and hence could not gain credit because such an

		<i>explanation can be applied to any situation. Also, some correctly stated that the current is periodic or that it changed continuously, except they did not elaborate how it is periodic and exactly in what way it is changing. (e.g. a constantly increasing current is changing continuously). There were also a lot of confusing and vague descriptions of the direction of the current, e.g., the current in the secondary coil being opposite to that in the primary coil, the coils are not connected so what exactly is the relationship between the direction in the coils?</i>	
		[Total: 6]	

7	(a)	The emission spectrum of atomic hydrogen consists of a number of discrete wavelengths	
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